

Part 1: Microprocessing Interfacing with Assembly Language
Lab 1: Introduction to ARM assembly language

Goals

1. Understanding of the fundamental concepts of Assembly Language
2. Able to write Simple Assembly program using ARM v7 ISA
3. Develop familiarity with a Computer System Simulator: Web based Cputator

Summary of Tasks:

1. Consult ARM Instruction Set Manual available at Canvas
2. Understand the following assembly example
 - a. The meaning of each of the identifiers in the program
 - b. How data segments and instruction segments work
3. Implement the following two program:
 - a. Write a simple assembly language program that is equivalent to if-else statement in C or Java
 - b. Write a simple assembly language program that is equivalent to *for/while/do-while loop* in C
 - c. Write a simple assembly language program that is equivalent to *procedure call (i.e. caller and callee functions)* in C
4. Do Something cool. The following are a few examples.
 - a. Implement simple algorithm such finding max or min in Assembly language
 - b. Implement switch-catch equivalent of C program in Assembly language

Assembly program example:

```
.org 1000 @program start at memory address 1000
.data
@put your data here
.text
.global _start

_start:
@ Initialize the sum to 0
mov r0, #0
b _start
```

Lab Demo

Lab 1: Introduction to ARM assembly language

Student Names: _____

Student IDs: _____

Time & Date: _____

Demo Part 1, 2, 3, Part 4 (optional) to lab instructor. And answer the following questions

1. Write about major categories of ARM instructions.

2. What are major Branch instructions in ARM

3. What does this ADDSEQ mean?

4. How CMP and BCond (i.e. BEQ, BNE) work together?